SECONDARY 1 (GRADE 7) 2023 CONTEST PAPER

NAME: _____ **Index Number:**

SCHOOL: _____

INSTRUCTIONS:

1. Please **DO NOT OPEN** the contest booklet until the Proctor has given permission to start.

2. TIME: **90 minutes**.

3. There are 35 questions:
   Section A: Questions 1 to 10 score 2 points each, no points are deducted for an unanswered question or wrong answer.
   Section B: Questions 11 to 30 score 3 points each, no points are deducted for an unanswered or wrong answer.
   Section C: Questions 31 to 35 score 4 points each, no points are deducted for an unanswered or wrong answer.

4. Shade your answers neatly using a **2B lead pencil** in the Answer Entry Sheet.

5. PROCTORING: No one may help any student in any way during the contest.

6. No electronic devices capable of storing and displaying visual information are allowed during the course of the exam.

7. **Calculators are allowed** into the exam.

8. All students must fill and shade their **Name, School and Index Number** in the Answer Entry Sheet and Contest booklet.

9. MINIMUM TIME: Students must stay in the exam hall for the full 90 minutes.

10. A student must show detailed working and transfer answers to the Answer Entry Sheet.

11. **No exam papers and written notes can be taken out by any contestant.**

Section A: 10 Open-ended Questions (10 X 2 marks = 20 marks)

1. Given the equation $20x - [4(y - 3x) + 9x] + 5x = m(sx + ty)$, $m > 3$ and m, s, and t are integers, find the sum of m, s, and t.

Answer: _____

2. Find the smallest possible integer m such that $120m$ is a perfect square.

Answer: _____

3. Consider a sequence where the nth term is given by $T_n = (3n + 2)^3$.

   Find the difference between the 6th and 5th term of the sequence.

Answer: _____

4. A sum of money was divided among Alice, Bob, and Carol in the ratio of $7:3:10$. If Bob received $\$80$ less than Carol, find the average amount of money received per person, in dollars. Round off the answer to the nearest integer.

Answer: _____

5. Emily, travelling at an average speed of $5\ km/h$, takes $40\ minutes$ to walk to the library. If she left her house for the library at $08:30\ AM$, calculate the average speed, in m/s, at which she must travel if she is to reach the library by $09:00\ AM$. Round off the answer to the nearest integer.

Answer: _____